

Image Classification Using Deep Learning: A Review

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Abstract

Image classification — the automatic assignment of a semantic label to a digital image — is one of the most studied problems in computer vision and a foundational application of deep learning. Prior to 2012, handcrafted feature descriptors combined with shallow classifiers dominated the field, yet their accuracy plateaued well below human-level performance. The advent of deep Convolutional Neural Networks (CNNs) fundamentally changed this landscape. This paper presents a narrative review of five landmark CNN architectures that defined the modern era of image classification: LeNet-5 (1998), AlexNet (2012), VGGNet (2015), ResNet (2016), and DenseNet (2017). For each architecture, we examine the structural design, training strategies, regularisation techniques, and benchmark performance on the ImageNet Large Scale Visual Recognition Challenge (ILSVRC). We then synthesise the findings to identify overarching trends — specifically the shift from shallow to deep networks, the resolution of the vanishing gradient problem through skip connections, and the move toward parameter-efficient designs. Our analysis shows that top-5 error rates on ImageNet declined from 15.3% (AlexNet, 2012) to 3.57% (ResNet-152, 2016) within four years — surpassing estimated human-level performance. We conclude with a discussion of open challenges and future directions, including lightweight mobile architectures, Vision Transformers, and self-supervised learning.

Index Terms — Deep Learning, Image Classification, Convolutional Neural Networks, AlexNet, VGGNet, ResNet, DenseNet, ImageNet, Transfer Learning.

I. INTRODUCTION

The ability to automatically assign a meaningful label to an image has long been a central goal of computer vision research. Image classification underpins a wide range of real-world systems: medical imaging platforms that detect tumours in radiographs, autonomous vehicles that identify pedestrians and road signs, agricultural systems that diagnose crop diseases from photographs, and surveillance infrastructure that recognises individuals in crowds. Despite its apparent simplicity as a problem statement, image classification remained extremely challenging for decades because images are high-dimensional, noisy, and subject to enormous variability in lighting, viewpoint, scale, and occlusion.

For most of the 1990s and 2000s, the dominant paradigm relied on hand-engineered feature extraction followed by a classical machine learning classifier. Algorithms such as Scale-Invariant Feature Transform (SIFT) [6] and Histogram of Oriented Gradients (HOG) [7] were designed to detect repeatable local patterns in images, which were then fed into classifiers such as Support Vector Machines (SVMs). While effective for constrained tasks, these pipelines suffered from a fundamental limitation: the features were engineered by human domain experts and therefore could

not adapt to the full complexity of natural images. Performance on large-scale benchmarks reached a ceiling that human-crafted features could not break through.

The situation changed dramatically in 2012, when Krizhevsky, Sutskever, and Hinton introduced AlexNet [1] — a deep Convolutional Neural Network trained end-to-end on raw pixel data using Graphics Processing Units (GPUs). AlexNet achieved a top-5 error rate of 15.3% on the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) 2012, reducing the error of the second-place entry by more than 10 percentage points. This result demonstrated conclusively that deep CNNs, given sufficient data and compute, could learn hierarchical feature representations far superior to any hand-engineered alternative. It triggered a wave of research into deeper, more powerful architectures that continues to this day.

This review paper traces that wave through four subsequent milestones. VGGNet [3] (2015) showed that very small convolution filters stacked to 16–19 layers could systematically improve accuracy. ResNet [4] (2016) introduced residual learning to enable networks of 50–152 layers, winning every track of ILSVRC 2015 and surpassing human-level performance. DenseNet [5] (2017) carried skip connectivity to its logical extreme by connecting every layer to all subsequent layers within a dense block, achieving competitive accuracy with dramatically fewer parameters. Together, these five architectures — anchored by the foundational work of LeCun et al. [2] who introduced the CNN itself — tell the story of how deep learning conquered image classification.

The primary objective of this review is to provide a structured, comparative analysis of these five architectures suitable for undergraduate CSE students who are encountering deep learning for the first time. The review addresses the following research questions:

RQ1. What are the key architectural innovations introduced by each landmark CNN model?

RQ2. How did these innovations address specific limitations of preceding architectures?

RQ3. What trends emerge when these architectures are compared on the ImageNet benchmark?

RQ4. What are the remaining challenges and future directions in image classification?

The remainder of this paper is organised as follows. Section II describes the search methodology used to identify and select the papers reviewed. Section III provides essential background on convolutional neural networks and the ImageNet benchmark. Section IV presents a review of each architecture. Section V provides a comparative analysis across all five models. Section VI discusses overarching findings and their implications. Section VII concludes the paper.

II. REVIEW METHODOLOGY

This review follows the narrative review methodology, which is appropriate when the goal is to synthesise and critically analyse a defined set of foundational works rather than to conduct an exhaustive systematic search of a research domain. The methodology consisted of three stages: source identification, selection, and quality assessment.

A. Source Identification

Literature was identified through searches of the following academic databases: Google Scholar (scholar.google.com), IEEE Xplore (ieeexplore.ieee.org), and the ACM Digital Library (dl.acm.org). Search terms included combinations of: "image classification", "convolutional neural network", "deep learning", "ImageNet", "AlexNet", "VGGNet", "ResNet", and "DenseNet". The initial search was not restricted by date. Additional papers were identified through forward and backward citation tracing from the primary sources.

B. Inclusion and Exclusion Criteria

Papers were included if they: (i) proposed an original CNN architecture for image classification; (ii) reported results on a standard benchmark dataset, particularly ImageNet ILSVRC; and (iii) were published in a peer-reviewed conference proceedings or journal. Papers were excluded if they: (i) addressed tasks other than image classification as the primary contribution; (ii) were review or survey papers themselves; or (iii) could not be accessed in full text. Applying these criteria to the initial candidate pool yielded the five primary papers reviewed in this work, selected on the basis of their foundational impact and high citation counts in the field.

C. Quality Assessment

Each selected paper was assessed against the following quality indicators: (i) number of citations (all five papers have received more than 10,000 citations each at time of writing); (ii) publication venue prestige (all were published at premier venues: NIPS, IEEE Xplore, ICLR, and CVPR); and (iii) reproducibility, as evidenced by public availability of code and pre-trained model weights. All five papers met these criteria at a high level. Table I in Section V presents a structured summary of the selected works.

III. BACKGROUND

A. Convolutional Neural Networks

A Convolutional Neural Network (CNN) is a class of deep learning model specifically designed to process grid-structured data such as images. Unlike fully connected networks, which apply a dense weight matrix to every input, CNNs exploit the spatial locality and translational invariance of images through three key operations: convolution, activation, and pooling.

In a convolutional layer, a set of learnable filters slides across the input feature map, computing a dot product at each spatial position to produce an activation map. Stacking multiple convolutional layers allows the network to build progressively more abstract representations: early layers detect

edges and textures, middle layers detect parts and shapes, and deeper layers detect high-level semantic concepts such as object categories. The Rectified Linear Unit (ReLU), defined as $f(x) = \max(0, x)$, is the standard activation function used between convolutional layers. ReLU introduces non-linearity while avoiding the vanishing gradient problem associated with sigmoid and tanh activations in deep networks. Pooling layers — typically max pooling — reduce the spatial dimensions of feature maps by retaining only the strongest activation within each local region, introducing a degree of spatial invariance and reducing computational cost. The final layers of a CNN are typically fully connected layers that map the extracted feature representations to a probability distribution over the target classes using a softmax function.

B. The ImageNet Benchmark

ImageNet is a large-scale image database assembled by Deng et al. containing over 14 million manually annotated images organised into more than 21,000 WordNet synsets. The ImageNet Large Scale Visual Recognition Challenge (ILSVRC), held annually from 2010 to 2017, used a standardised subset of approximately 1.2 million training images, 50,000 validation images, and 100,000 test images across 1,000 object categories. The primary evaluation metric is top-5 error rate: a prediction is counted as correct if the ground-truth class appears among the model's five most confident predictions. ILSVRC served as the definitive benchmark driving architectural innovation throughout the period covered by this review.

C. The Vanishing Gradient Problem

A critical challenge in training very deep networks is the vanishing gradient problem. During backpropagation, gradients are computed by the chain rule and multiplied together across every layer. In networks with many layers and non-saturating activation functions, these multiplied gradients become exponentially small as they propagate toward the earlier layers, effectively preventing those layers from learning. This problem constrained practical network depth to fewer than approximately ten layers for most of the 2000s. The introduction of ReLU activations partially mitigated the problem for moderate depths, but did not fully resolve it for very deep networks. The residual learning framework introduced by ResNet [4] provided a principled architectural solution, as discussed in Section IV-D.

IV. REVIEW OF CNN ARCHITECTURES FOR IMAGE CLASSIFICATION

A. LeNet-5 (LeCun et al., 1998)

LeNet-5, proposed by LeCun, Bottou, Bengio, and Haffner [2], stands as the direct precursor of all modern CNN architectures. Designed for handwritten digit recognition on the MNIST dataset, LeNet-5 introduced the core design pattern — convolutional layers followed by subsampling layers followed by fully connected layers — that remains central to image classification networks today.

Architecture

LeNet-5 consists of seven layers: two convolutional layers (C1, C3), two average-pooling subsampling layers (S2, S4), and three fully connected layers (C5, F6, Output). The input is a 32×32 greyscale image. The first convolutional layer applies six 5×5 filters, producing six 28×28 feature maps. After subsampling to 14×14 , a second convolutional layer applies sixteen 5×5 filters, followed by a second subsampling to 5×5 . The network uses the tanh activation function throughout and contains approximately 60,000 parameters in total.

Results and Contribution

LeNet-5 achieved an error rate of approximately 0.95% on the MNIST test set, which was state-of-the-art at the time of publication. Its key contribution was demonstrating empirically that hierarchical feature representations could be learned directly from raw pixel data via gradient descent, without requiring hand-designed features. This proof-of-concept was foundational: every CNN reviewed in subsequent sections can be understood as a direct architectural descendant of LeNet-5.

Limitations

The primary limitation of LeNet-5 was its inability to scale to larger images and more complex tasks. The network was designed for 32×32 single-channel inputs with only 10 output classes. Hardware limitations of the 1990s — specifically the absence of GPU-accelerated computation and large-scale labelled datasets — prevented researchers from exploring deeper or wider variants. The architectural insights of LeNet-5 thus lay largely dormant for over a decade until the conditions for scaling them arrived.

B. AlexNet (Krizhevsky, Sutskever & Hinton, 2012)

AlexNet [1] marks the beginning of the deep learning era in computer vision. Introduced at the Neural Information Processing Systems conference in 2012, it won the ILSVRC competition by an unprecedented margin and triggered the explosive growth in deep learning research that followed.

Architecture

AlexNet comprises eight learned layers: five convolutional layers and three fully connected layers, totalling approximately 60 million parameters and 650,000 neurons. The first convolutional layer applies 96 large 11×11 filters with a stride of 4 to a $224 \times 224 \times 3$ colour input. Subsequent convolutional layers use smaller 5×5 and 3×3 filters with varying numbers of feature maps (256, 384, 384, and 256 respectively). The three fully connected layers have 4,096, 4,096, and 1,000 neurons, with a final softmax for class probabilities. The network was distributed across two NVIDIA GTX 580 GPUs, with the split between GPUs occurring at specific layers.

Key Innovations

AlexNet introduced four innovations that proved critical to its success and became standard practice in subsequent architectures. First, it adopted the Rectified Linear Unit (ReLU) activation function in place of tanh, which the authors demonstrated trained six times faster on equivalent networks. Second, it employed Dropout regularisation [8] with probability $p = 0.5$ in the two

largest fully connected layers, effectively training an ensemble of sub-networks and dramatically reducing overfitting. Third, it applied local response normalisation, inspired by lateral inhibition in neuroscience. Fourth, extensive data augmentation — random crops, horizontal reflections, and principal component colour jitter — artificially increased the effective training set size from 1.2 million to far larger.

Results

AlexNet achieved a top-5 error rate of 15.3% on ILSVRC 2012, compared to 26.2% for the second-place entry. This reduction of more than 10 percentage points was unprecedented in the history of the competition and demonstrated conclusively that deep CNNs trained on GPUs could outperform all prior computer vision approaches by a substantial margin.

C. VGGNet (Simonyan & Zisserman, 2015)

VGGNet, developed at the University of Oxford's Visual Geometry Group and published by Simonyan and Zisserman [3], investigated a systematic question: given a fixed architectural template, how does network depth affect classification accuracy?

Architecture

The defining characteristic of VGGNet is its use of exclusively 3×3 convolutional filters, the smallest possible size that captures spatial context, stacked in sequences of 2–4 layers between max-pooling operations. This design principle produces a uniform, highly regular architecture. Two configurations received the most attention: VGGNet-16 (13 convolutional + 3 fully connected layers) and VGGNet-19 (16 convolutional + 3 fully connected layers). All convolutional layers use a stride of 1 and same-padding, so that spatial dimensions are preserved within each convolutional block and halved only by the five max-pooling operations.

Rationale for 3×3 Filters

The choice of 3×3 filters requires justification, as AlexNet had used larger 11×11 and 5×5 filters in its earlier layers. Simonyan and Zisserman showed that two stacked 3×3 convolutional layers have the same effective receptive field as a single 5×5 layer, and three stacked 3×3 layers have the same receptive field as a single 7×7 layer. Crucially, the stacked smaller filters have fewer parameters and introduce more non-linearities, both of which are beneficial: fewer parameters reduce overfitting risk, and more non-linearities increase the representational capacity of the network.

Results and Limitations

VGGNet-16 achieved a top-5 error rate of 7.3% on ILSVRC 2014, securing first place in the localisation task and second place in the classification task. Its architectural simplicity made it extensively adopted as a feature extractor for transfer learning. However, VGGNet's principal limitation is its parameter count: VGGNet-16 contains approximately 138 million parameters, almost entirely concentrated in the three fully connected layers, making it expensive to store, train from scratch, and deploy on resource-constrained devices.

D. ResNet (He, Zhang, Ren & Sun, 2016)

ResNet, developed at Microsoft Research Asia and published by He et al. [4], addressed the fundamental obstacle to making networks deeper: the degradation problem. Counterintuitively, adding more layers to a sufficiently deep plain network causes both training and validation accuracy to decrease — not due to overfitting, but due to optimisation difficulty.

Residual Learning

The key insight of ResNet is the residual learning framework. Instead of requiring each stack of layers to learn a direct mapping $H(x)$ from input x to output, ResNet reformulates the problem as learning a residual function $F(x) = H(x) - x$, so that the layer output becomes $F(x) + x$. This is implemented through a shortcut connection — also called a skip connection — that adds the identity mapping of the layer input directly to its output, bypassing the nonlinear transformations. Crucially, if the optimal transformation is close to the identity (which is often the case when a plain network has already reached near-optimal depth), the residual function $F(x)$ merely needs to push toward zero, which is far easier to learn than an arbitrary transformation. Shortcut connections add no parameters and negligible computation.

Architecture

ResNet was evaluated in variants of 18, 34, 50, 101, and 152 layers. The deeper variants (50+) use a bottleneck building block: a 1×1 convolution to reduce dimensionality, a 3×3 convolution, and a 1×1 convolution to restore dimensionality, which reduces the number of parameters required per block. ResNet-152 has 152 layers — eight times deeper than VGGNet — yet has lower computational complexity because the bottleneck design keeps the parameter count manageable at approximately 60 million.

Results

An ensemble of ResNet models achieved a top-5 error rate of 3.57% on ILSVRC 2015, winning first place on all five competition tracks: image classification, single-object localisation, object detection, object detection from video, and scene classification. Notably, a single ResNet-152 model achieved 4.49% top-5 error, which is below the estimated human-level error rate of 5.1% on the same benchmark — the first time a machine learning system had matched human performance on this task.

E. DenseNet (Huang, Liu, van der Maaten & Weinberger, 2017)

DenseNet, introduced by Huang et al. [5] and awarded Best Paper at CVPR 2017, extended the principle of skip connections to its logical extreme by establishing direct connections between every layer and all subsequent layers within a dense block.

Dense Connectivity

In a standard feedforward network with L layers, there are L connections (one per layer). In a ResNet, an additional L shortcut connections are added. In a DenseNet with L layers, there are

$L(L+1)/2$ direct connections. Formally, the output of the l -th layer is: $x_l = H_l([x_0, x_1, \dots, x_{l-1}])$, where $[\cdot]$ denotes feature map concatenation and H_l is a composite function of Batch Normalisation, ReLU, and a 3×3 convolution. This dense connectivity has three consequences. First, it maximises feature reuse: features computed by early layers remain directly accessible to all later layers, eliminating the need to recompute them. Second, it strengthens gradient flow: every layer has direct access to the loss gradient through multiple short paths, alleviating the vanishing gradient problem without any additional mechanisms. Third, it acts as a form of implicit deep supervision: each layer receives direct supervision from the loss function.

Architecture and Results

DenseNet is organised into dense blocks separated by transition layers (1×1 convolution followed by 2×2 average pooling) that reduce spatial dimensions and control the number of feature maps. A growth rate hyperparameter k controls how many new feature maps each layer contributes.

DenseNet-121 ($k=32$) achieved a top-5 error rate of 5.35% on ImageNet, competitive with ResNet-50 (5.25%) while using less than one-third the parameters (8 million versus 25 million).

DenseNet has found particularly wide adoption in medical image analysis, where training datasets are small and the aggressive feature reuse of dense connectivity acts as an effective regulariser.

V. COMPARATIVE ANALYSIS

Table I summarises the five reviewed architectures across the dimensions of depth, ImageNet top-5 error rate, number of parameters, and primary architectural innovation. The data in the table enables three cross-cutting observations, each addressing one of the research questions posed in Section I.

TABLE I

Comparison of CNN Architectures for Image Classification

Model	Year	Layers	Top-5 Error	Key Innovation	Parameters
LeNet-5	1998	7	~0.95% *	First CNN for images	~60K
AlexNet	2012	8	15.3%	ReLU, Dropout, GPU training	60M
VGGNet-16	2015	16	7.3%	Stacked 3×3 conv filters	138M
ResNet-50	2016	50	5.25%	Residual skip connections	25M
DenseNet-121	2017	121	5.35%	Dense feature connectivity	8M

* LeNet-5 performance on MNIST (10 classes, greyscale). All other values are top-5 error on ImageNet ILSVRC (1,000 classes, colour).

A. Increasing Depth Correlates with Improved Accuracy (RQ1 & RQ3)

The most visually apparent trend in Table I is the systematic reduction in top-5 error rate as architectures grew deeper. From AlexNet's 15.3% (8 layers) to VGGNet's 7.3% (16 layers), then ResNet's 5.25% (50 layers) and beyond, increasing depth consistently delivered accuracy gains. However, this trend does not hold unconditionally: plain networks (without skip connections) degrade when made too deep, as observed in experiments by He et al. [4] comparing plain 34-layer and 56-layer networks against their residual counterparts. Depth improves accuracy only when paired with appropriate mechanisms for gradient propagation — specifically, residual connections in ResNet and dense connections in DenseNet.

B. Each Architecture Solved the Bottleneck of Its Predecessor (RQ2)

A central finding of this review is that the architectural evolution was not incremental but reactive: each major architecture directly addressed the dominant limitation of the previous generation. AlexNet overcame the data-scarcity and computational constraints that had prevented LeNet-style networks from scaling, through GPUs and large datasets. VGGNet overcame the arbitrariness of AlexNet's large-filter design by demonstrating that a principled all-small-filter architecture could be more effective and more uniform. ResNet overcame the degradation problem that had prevented VGGNet-style architectures from being made substantially deeper, through the residual shortcut. DenseNet addressed the parameter inefficiency of ResNet's additive shortcuts by replacing addition with concatenation, enabling aggressive feature reuse.

C. Architectural Innovation Matters More Than Parameter Count (RQ3)

Perhaps the most practically important trend in Table I is the decoupling of accuracy from parameter count. VGGNet-16 has 138 million parameters — more than double AlexNet's 60 million — yet achieves substantially better accuracy. ResNet-50, with only 25 million parameters, outperforms VGGNet-16 on both accuracy and efficiency. DenseNet-121 achieves similar accuracy to ResNet-50 with only 8 million parameters. This progression strongly suggests that architectural innovation — the design of how information flows through the network — is a more powerful driver of accuracy improvement than simply adding more parameters. The lesson for practitioners is that careful architectural design can substitute for raw computational scale.

VI. DISCUSSION

A. The Role of Transfer Learning

One of the most practically significant consequences of the architectures reviewed in this paper is their role as pre-trained feature extractors in transfer learning. A CNN pre-trained on the 1.2 million images of ImageNet develops rich, general-purpose visual representations in its convolutional layers that transfer remarkably well to entirely different visual domains. Fine-tuning a pre-trained ResNet-50 on a small medical imaging dataset, for instance, consistently outperforms training a network of the same architecture from random initialisation, even when the target domain bears little visual resemblance to ImageNet. This transfer learning paradigm has dramatically reduced the data and computational requirements for deploying deep learning in

specialised applications and is one reason why the architectures reviewed in this paper remain in active use a decade after their introduction.

B. Limitations of the Reviewed Architectures

Despite their considerable achievements, all five architectures share important limitations that motivated subsequent research directions. First, they are computationally expensive at inference time and are therefore difficult to deploy on mobile or embedded devices. ResNet-50 requires approximately 4 billion floating-point operations (FLOPs) per image, which is feasible on a modern server GPU but prohibitive on a smartphone processor. This limitation motivated the development of lightweight architectures such as MobileNet [9] and EfficientNet, which use depthwise separable convolutions and neural architecture search, respectively, to achieve competitive accuracy at a fraction of the computational cost.

Second, all five architectures rely on large quantities of labelled training data. ImageNet required over 25,000 human annotators to produce its 1.2 million labels — an effort that cannot be replicated for every application domain. This dependency has motivated research into self-supervised and semi-supervised learning approaches that can leverage unlabelled data. Third, CNNs are largely opaque: it is difficult to explain why a particular prediction was made, which is a significant barrier to adoption in safety-critical domains such as medical diagnosis and autonomous driving.

C. Emerging Directions

Several research directions have emerged in response to the limitations described above. Vision Transformers (ViT) [10], introduced in 2020, demonstrated that the self-attention mechanism from natural language processing could replace convolutional layers entirely, achieving state-of-the-art accuracy on ImageNet when trained on sufficiently large datasets. Subsequent work has explored hybrid CNN-Transformer architectures that capture the local inductive biases of convolutions alongside the global context modelling of attention. Self-supervised learning methods, such as contrastive learning (SimCLR, MoCo) and masked image modelling (MAE), have shown that powerful visual representations can be learned from unlabelled images, dramatically reducing the need for large annotated datasets. These directions suggest that while the convolutional architectures reviewed in this paper remain important reference points, the frontier of image classification research has moved significantly beyond them.

VII. CONCLUSION

This paper has reviewed five landmark deep learning architectures for image classification — LeNet-5, AlexNet, VGGNet, ResNet, and DenseNet — tracing the evolution of the field from the first proof-of-concept CNN for digit recognition to networks that surpass human-level performance on a large-scale natural image benchmark.

In response to the research questions posed in Section I: the key architectural innovations were, in chronological order, the end-to-end learning of hierarchical features from raw pixels (LeNet-5), GPU-accelerated training with ReLU and Dropout (AlexNet), systematic depth via uniform 3×3 filters (VGGNet), residual skip connections for ultra-deep optimisation (ResNet), and dense connectivity for maximal feature reuse (DenseNet) (RQ1). Each innovation directly resolved the dominant bottleneck of the preceding architecture (RQ2). The overarching trend is one of steadily declining error rates, achieved not through linear scaling of existing designs but through qualitative architectural breakthroughs that improved gradient flow, feature reuse, and parameter efficiency simultaneously (RQ3). Open challenges include computational efficiency for mobile deployment, reduced dependence on labelled data, and improved interpretability of learned representations (RQ4).

For CSE students, the practical takeaway is that these five architectures, together with their pre-trained ImageNet weights, represent freely available, highly capable starting points for a vast range of computer vision applications through transfer learning. Understanding their design principles provides the conceptual foundation necessary for engaging with the current state of the art, including Vision Transformers and self-supervised learning, which build directly on the insights established by the architectures reviewed here.

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